11. Comparing PE and WPE classification performance on raw datasets

Below is a direct comparison between standard Shannon Permutation Entropy (PE) and Weighted Permutation Entropy (WPE) across all datasets for both within-dataset kmeans clustering and cross-dataset centroid evaluation.

11.1. Metric-by-Metric Performance Comparison: PE vs. WPE

The following tables present a detailed, metric-by-metric comparison between standard Shannon Permutation Entropy (PE) and Weighted Permutation Entropy (WPE), covering both within-dataset k-means clustering and cross-dataset centroid transfer evaluations.

11.1.1. Balanced Accuracy Comparison

Target Group	Evaluation Context / Centroids	PE Acc.	WPE Acc.	Acc. Δ
Within-Dataset Clustering (Self-Trained Centroids)				
DBT2/NDBT2	K-Means (Self)	0.98	0.97	−0.01
CGMacros + Dexcom	K-Means (Self)	0.79	0.86	**+0.07**
CGMacros + Libre	K-Means (Self)	0.90	0.86	−0.04
Average CGMacros	K-Means (Self)	0.845	0.86	+0.015
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.93	0.91	−0.02
Hall	K-Means (Self)	0.85	0.79	−0.06
Cross-Dataset Centroid Validation Transfer				
CGMacros + Dexcom	DBT2/NDBT2	0.83	0.86	+0.03
CGMacros + Libre	DBT2/NDBT2	0.93	0.90	−0.03
Hall	DBT2/NDBT2 + CGMacros	0.86	0.75	**−0.11**

Table 21: Balanced Accuracy comparison between PE (HxC) and WPE (WHxC) feature spaces.

11.1.2. F1-Score Comparison

Target Group	Evaluation Context / Centroids	PE F1	WPE F1	F1 Δ
Within-Dataset Clustering (Self-Trained Centroids)				
DBT2/NDBT2	K-Means (Self)	0.98	0.98	0.00
CGMacros + Dexcom	K-Means (Self)	0.81	0.86	**+0.05**
CGMacros + Libre	K-Means (Self)	0.90	0.86	−0.04
Average CGMacros	K-Means (Self)	0.855	0.86	+0.005
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.93	0.91	−0.02
Hall	K-Means (Self)	0.93	0.87	−0.06
Cross-Dataset Centroid Validation Transfer				
CGMacros + Dexcom	DBT2/NDBT2	0.81	0.86	+0.05
CGMacros + Libre	DBT2/NDBT2	0.93	0.89	−0.04
Hall	DBT2/NDBT2 + CGMacros	0.95	0.92	−0.03

Table 22: F1-Score performance comparison between PE and WPE.

11.2. Sensitivity Comparison

Target Group	Evaluation Context / Centroids	PE Sens.	WPE Sens.	Sens. Δ
Within-Dataset Clustering (Self-Trained Centroids)				
DBT2/NDBT2	K-Means (Self)	0.95	1.00	+0.05
CGMacros + Dexcom	K-Means (Self)	0.87	0.80	−0.07
CGMacros + Libre	K-Means (Self)	0.87	0.80	−0.07
Average CGMacros	K-Means (Self)	0.87	0.80	−0.07
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.92	0.88	−0.04
Hall	K-Means (Self)	0.89	0.79	**−0.10**
Cross-Dataset Centroid Validation Transfer				
CGMacros + Dexcom	DBT2/NDBT2	0.73	0.80	+0.07
CGMacros + Libre	DBT2/NDBT2	0.87	0.80	−0.07
Hall	DBT2/NDBT2 + CGMacros	0.92	0.89	−0.03

Table 23: Sensitivity (True Positive Rate) comparison between PE and WPE.

11.3. Specificity Comparison

Target Group	Evaluation Context / Centroids	PE Spec.	WPE Spec.	Spec. Δ
Within-Dataset Clustering (Self-Trained Centroids)				
DBT2/NDBT2	K-Means (Self)	0.91	0.95	+0.04
CGMacros + Dexcom	K-Means (Self)	0.71	0.93	**+0.22**
CGMacros + Libre	K-Means (Self)	0.93	0.93	0.00
Average CGMacros	K-Means (Self)	0.82	0.93	**+0.11**
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.94	0.94	0.00
Hall	K-Means (Self)	0.80	0.80	0.00
Cross-Dataset Centroid Validation Transfer				
CGMacros + Dexcom	DBT2/NDBT2	0.93	0.93	0.00
CGMacros + Libre	DBT2/NDBT2	1.00	1.00	0.00
Hall	DBT2/NDBT2 + CGMacros	0.80	0.60	**−0.20**

Table 24: Specificity (True Negative Rate) comparison between PE and WPE.

Result : Key Analytical Insights per Metric:

- **Balanced Accuracy**: WPE shows significant improvements on high-variance CGM signals (Dexcom: +0.07 self, +0.03 transfer), while regular PE performs better on smoother datasets (Hall: −0.06 self, −0.11 transfer).
- **F1-Score**: WPE matches or exceeds PE on Dexcom (+0.05), maintaining overall competitive F1-scores across diabetic populations.
- **Sensitivity**: Standard PE maintains a generally higher sensitivity across most groups, whereas WPE trades some sensitivity for lower false-alarm rates.
- **Specificity**: WPE's primary advantage is in specificity on noisy devices, showing an outstanding +0.22 gain on Dexcom in self-clustering and maintaining 1.00 on Libre centroid transfer.